# Supplementary Note

We searched the literature for previous associations between polymorphic genes in QTL hot spots and their linked SMPs, or pathways modulated by these linked SMPs. For candidate selection we only considered genes located in the confidence intervals that contain coding-region, as opposed to non-coding, polymorphisms. The strongest literature-supported associations are discussed below for two QTL hot spots. We stress that what follows are associations which must ultimately be verified.

For the QTL hot spot located on chromosome 1, there are two polymorphic genes: *YAT1* (*YAR035W*) and *SWH1* (*YAR042W*). *YAT1* encodes an outer mitochondrial carnitine acetyltransferase, and is polymorphic at two residues [1]. *SWH1* encodes a protein with sequence homology to mammalian oxysterol-binding protein, and is polymorphic at 14 residues [2]. We favor *SWH1* and disfavor *YAT1* as a candidate for two reasons. First, carnitine contains a quaternary ammonium (quat) center, and none of the SMPs linking to this hot spot do. In fact, the classic quat-containing compound, benzalkonium, is present in our data set but links to the QTL hot spot on chromosome 12, suggesting that the *YAT1* polymorphisms may be neutral. Second, several of the SMPs linking to the QTL hot spot on chromosome 1 – ketoconazole and haloperidol – modulate sterol biosynthesis in yeast, and mutations at *SWH1* cause pleiotropic sterol-related phenotypes [3].

For the QTL hot spots on chromosome 14, there are numerous polymorphic genes, including ones previously shown to affect two disparate quantitative traits in yeast [4,5]. The TOR kinase inhibitor rapamycin, among others, links here. For several reasons, we favor *MKS1* (*YNL076W*), which encodes a pleiotropic regulatory factor downstream of the TOR kinases [6]. First, the *mks1* deletion mutant displays greater resistance to rapamycin [7]. Second, *MKS1* is polymorphic at a single residue that is located in a conserved domain of the protein [8]. Finally, the BY allele of *MKS1* is the derived allele, in a situation reminiscent of and analogous to *PHO84*, which we verified as the resistance-conferring polymorphism.

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